

HARMILAP SINGH DHALIWAL

Machine Learning Engineer · Text, Search & Applied ML · Production Systems from Concept to Delivery

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Machine learning engineer with 8+ years in applied ML and a record of shipping text-centric systems from dataset and architecture design through deployment and operation. Built three production AI systems in regulated finance spanning agentic workflows, LLM analytics, semantic retrieval, ranking, and guarded text-to-SQL. Earlier work includes document intelligence, classical ML, deep learning, and an industrial digital twin. Remain hands-on across ~70% of delivery while leading cross-functional production work and mentoring engineers and interns.

PROFESSIONAL EXPERIENCE

AI & Data Science Lead, Finance | Royal Bank of Canada

April 2025 - Present

CFO Group, Enterprise Finance · Toronto, ON

Hands-on across architecture, modeling, and full-stack implementation while leading cross-functional productionisation; mentor one Senior AI Scientist and interns (9 cumulative, peak 5) and contribute to AI/ML hiring.

Production AI Systems

- › Drove PAR Assist from concept to production, the first true agentic AI platform approved for production at the bank. Built the core application stack: ETL, a single-agent LangGraph orchestrator, LLM-driven MCP tool sub-routines, PostgreSQL with pgvector dense retrieval, document ingestion, and the authoring workflow. Pilot in April 2026; full CFO Group launch across all geographies in May 2026.
- › Built a multi-agent successor now in pilot.
- › Conceived, architected, and built Astraeus, a multi-stage LLM analytics platform for governed compensation-cost queries across roughly 40,000 cost centres; led its cross-functional productionisation. The LLM routes intent while entitlement-aware deterministic Python performs retrieval and calculation, limiting inputs sent to approved model endpoints.
- › Led a focused two-week concurrent refactor of Aegis v1 into v2 with the production team: dense semantic retrieval surfaces candidate metrics, LLM-assisted disambiguation resolves near-duplicate names, and reviewed templates plus parameter binding constrain text-to-SQL execution.

Technical Leadership

- › Mentor through architecture and code reviews on retrieval, model behavior, and SQL guardrails; established review and refactoring practices in GitHub that helped junior contributors ship production code.
- › Prepare LLM-as-judge evaluation evidence and extensive human-test results for Model Risk review, supported by custom tracing and logging; remain accountable for application behavior and post-release follow-through.

Enterprise Finance Senior Data Scientist | Royal Bank of Canada

September 2022 - April 2025

CFO Group, Enterprise Finance · Toronto, ON

- › Built Aegis v1 end-to-end: automated text extraction and entity matching over Big 6 Canadian bank Supplementary Financial Packages whose schemas shift quarterly, removing a long-standing bottleneck in peer analysis. CFO One RBC Team Award (2025).
- › Reduced commodity-tax return preparation from months to 90 minutes with a PySpark pipeline supporting allocation of roughly \$600M across enterprise general-ledger data. CFO Group RBC Quarterly Team Award (Q4 2023).
- › Built finance-automation pipelines in Python and PySpark with Tableau reporting, including planned-versus-actual and journal-entry preparation workflows adopted across the finance team.

EARLIER EXPERIENCE

Machine Learning Engineer | Quantiphi Inc.

October 2021 - September 2022

Toronto, ON

- › Improved checkbox-detection accuracy from roughly 70% to 99.95% on Humana claims documents by layering OpenCV pixel-level detection and Random Forest classification over OCR.
- › Built document-classification and entity-extraction workflows and deployed custom and AutoML models on Vertex AI; developed supporting data infrastructure with Bigtable and BigQuery.

Data Scientist | Tata Consultancy Services

August 2016 - November 2019

Pune, India

- › Delivered \$3M in annual savings for a 900 MW generating unit at Kansai Electric’s 1,800 MW Maizuru coal-fired station in Japan. As sole data scientist on a three-person R&D team, built the combustion-tuning digital twin end-to-end: ETL across 90+ sensors, 84 regression models, and Particle Swarm Optimization that produced operator-reviewed boiler-tuning recommendations.
- › Designed the model-selection framework using k-fold cross-validation, R², RMSE, MAPE, and fold variance; also shipped LSTM forecasting and XGBoost transformer-life prediction models.
- › Placed 2nd of 600 in TCS’s computer-vision hackathon with a CNN math-notation detector; earned the Innovation Pride Award (2019).

TECHNICAL SKILLS

Machine Learning	PyTorch, TensorFlow, scikit-learn, XGBoost; regression, classification, clustering, time series, CNNs, Transformers; model selection and evaluation
Text, Search & Ranking	Document classification and extraction, named entities, embeddings, semantic retrieval and ranking, near-duplicate disambiguation, dense RAG, guarded text-to-SQL
Agentic Systems	LangGraph, LangChain, MCP, prompt and context design, multi-stage LLM pipelines, human-in-the-loop workflows, deterministic orchestration, LLM evaluation
Production & Data	Python, SQL, R, PySpark, ETL, PostgreSQL/pgvector, OpenShift CI/CD, observability, Model Risk review
Cloud	Google Cloud Platform, Vertex AI, AutoML, BigQuery, Bigtable

EDUCATION & CREDENTIALS

Post Graduate Professional Certificate, Big Data Analytics

Georgian College · 2021

B.Eng., Electronics & Communications Engineering

Thapar University · 2016

Certifications: DeepLearning.AI Deep Learning; TensorFlow Developer; Johns Hopkins Data Science specializations

SELECTED RECOGNITION

- › CFO One RBC Team Award (2025): Aegis v1 productionisation
- › CFO Group RBC Quarterly Team Award (Q4 2023): Commodity Tax automation
- › Innovation Pride Award, TCS (2019): 2nd of 600 in computer-vision hackathon